

/ Agenda & Speakers

- **Intro** – Keaton Andersen
- **PSoC6 AI Kit Overview** – Keaton Andersen
- **Utilizing Sensors with /IOTCONNECT** – Keaton Andersen
- **DEEPCRAFT™ Overview** – Lukas Wirkestrand
- **DEEPCRAFT™ Ready Models** – Lukas Wirkestrand
- **DEEPCRAFT™ Ready Models Showcase** – Steven Dettloff
 - IMU-Based Models (Fall Detection – Elderly Care)
 - Microphone-Based Models (Smart Manufacturing – Siren & Alarm Detection)
 - Radar-Based Models (Gesture Detection & Retail Scenarios)
- **Q&A + Resources** - All



Keaton Andersen
Sr Manager, Customer Solutions
Avnet



Lukas Wirkestrand
System Applications Engineer
Imagimob



Steven Dettloff
IoT Solutions Manager
Avnet

Product Introduction: PSOC™ 6 AI Kit

Introduction to PSOC™ 6 AI Evaluation Kit and key features.

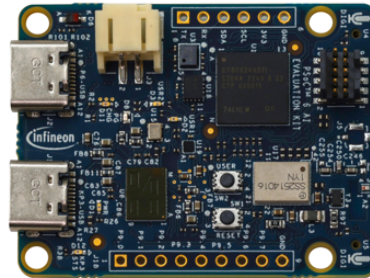
Sensor overview (IMU, radar, microphones, pressure) & technical specs.

Featured Ready Model Applications

Factory Alarm



Infineon PSOC™ 6 Artificial Intelligence Evaluation Kit



/IOTCONNECT™



Fall Detection

Gesture Recognition

Featured Ready Model Applications

Factory Alarm

Fall Detection

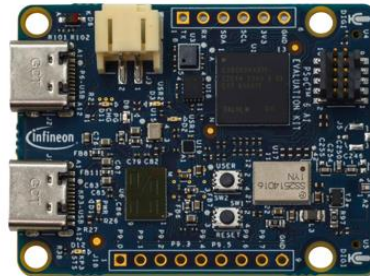
Gesture Recognition



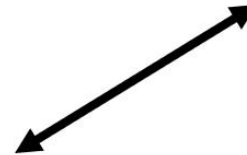
Fall Detection
Ready Model



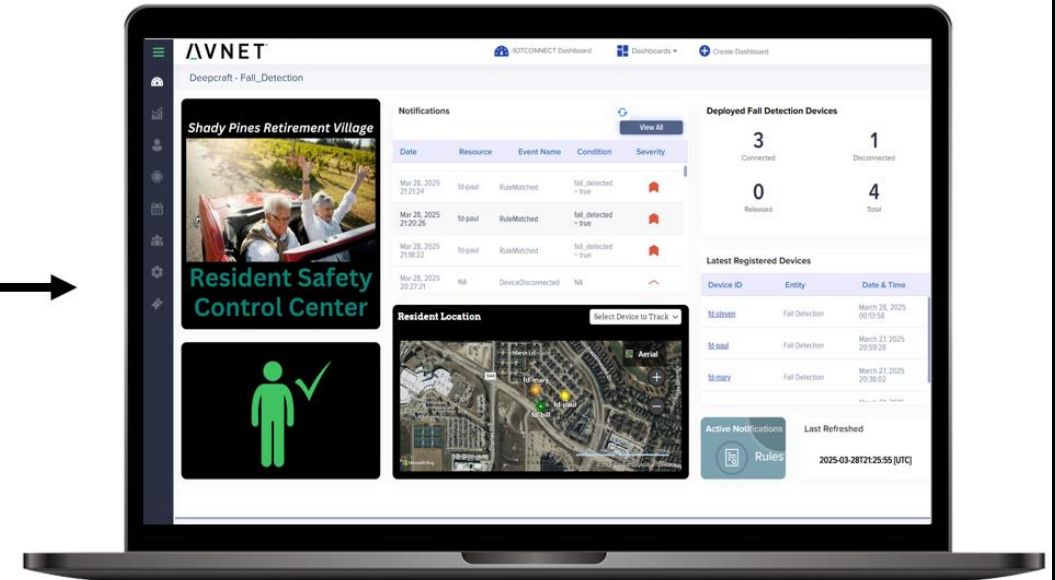
Infineon PSOC™ 6 Artificial Intelligence
Evaluation Kit



Onboard
IMU



/IOTCONNECT™



Featured Ready Model Applications

Factory Alarm

Fall Detection

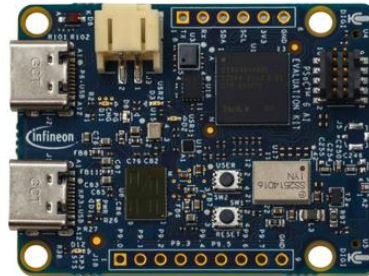
Gesture Recognition



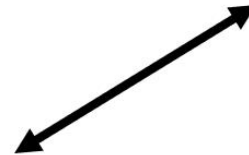
Gesture Recognition
Ready Model



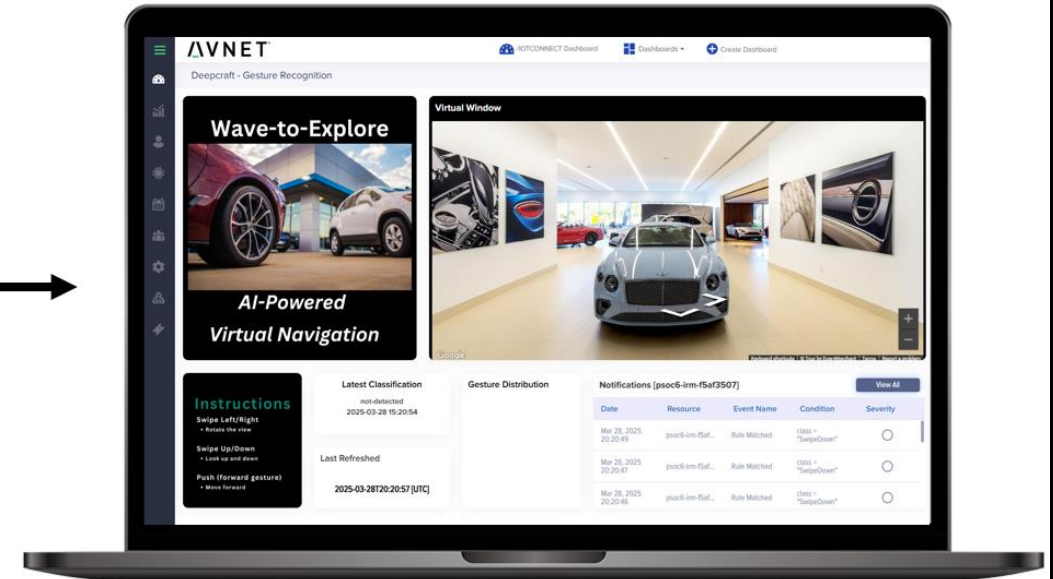
Infineon PSoC™ 6 Artificial Intelligence
Evaluation Kit



Onboard
Radar Sensor



/IOTCONNECT™



/ PSOC™ 6 AI Evaluation Kit

Infineon's HW Platform for Edge AI

www.infineon.com/aikit



Easy to use & low-cost

Evaluation board with efficient form factor for easy prototyping



Easy creation of ML models

Sensor fusion-, ML-, acoustic-, time series- and radar models



Connect easily & continuously improve models

Wired & wirelessly



CY8CKIT-062S2-AI

Scan the QR code for more



Fast time to market at minimal development cost



Direct launch for PSOC™ 6 & AURIX™

End-to-end



Collect data, create, train, evaluate & deploy your ML models fast



The best of Hardware, Software & ML



Solution approach with PSOC™ 6, QSPI Flash & multi-sensor input: radar, microphone, pressure & 6-axis motion sensor

PSOC™ 6

High-performance dual-core architecture for the IoT

Most flexible, high-performance, dual-core architecture designed specifically for the IoT:

- High performance 150-MHz and 100-MHz dual-core Arm® Cortex®-M4 and Arm® Cortex®-M0+ architecture
- Ultra-low-power 40-nm technology and design that consumes as little as 22-μA/MHz in active power mode
- High-performance dual-core architecture to optimize system power consumption and performance

PSOC™ 6 high-performance, dual-core MCU architecture

**Cortex®-M4
usage examples:**

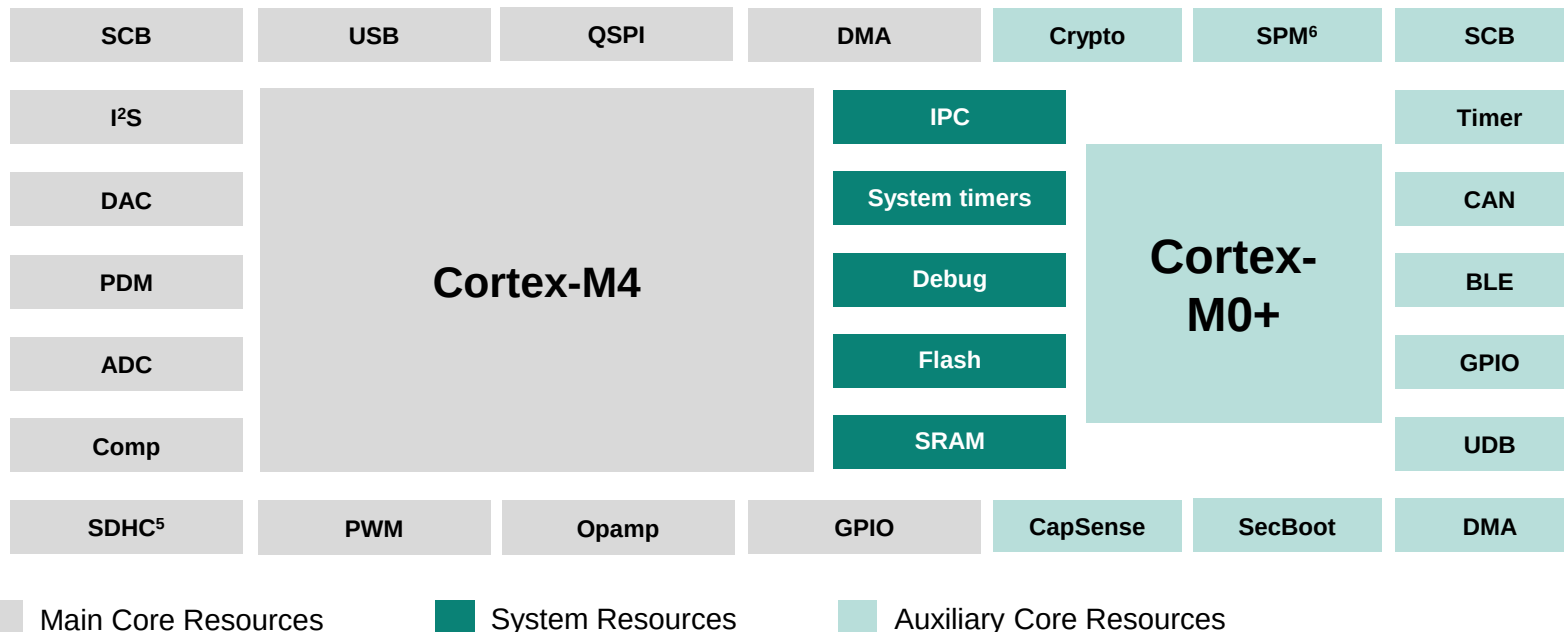
RTOS

Displays

Sensor analytics

Audio interface

USB/BLE HCI



**Cortex®-M0+
usage examples:**

Wireless stack

CapSense

Secure functions

I/O data control

Sensor aggregation

/ AI evaluation kit software architecture

Application examples

Imagimob data collection
and deployment

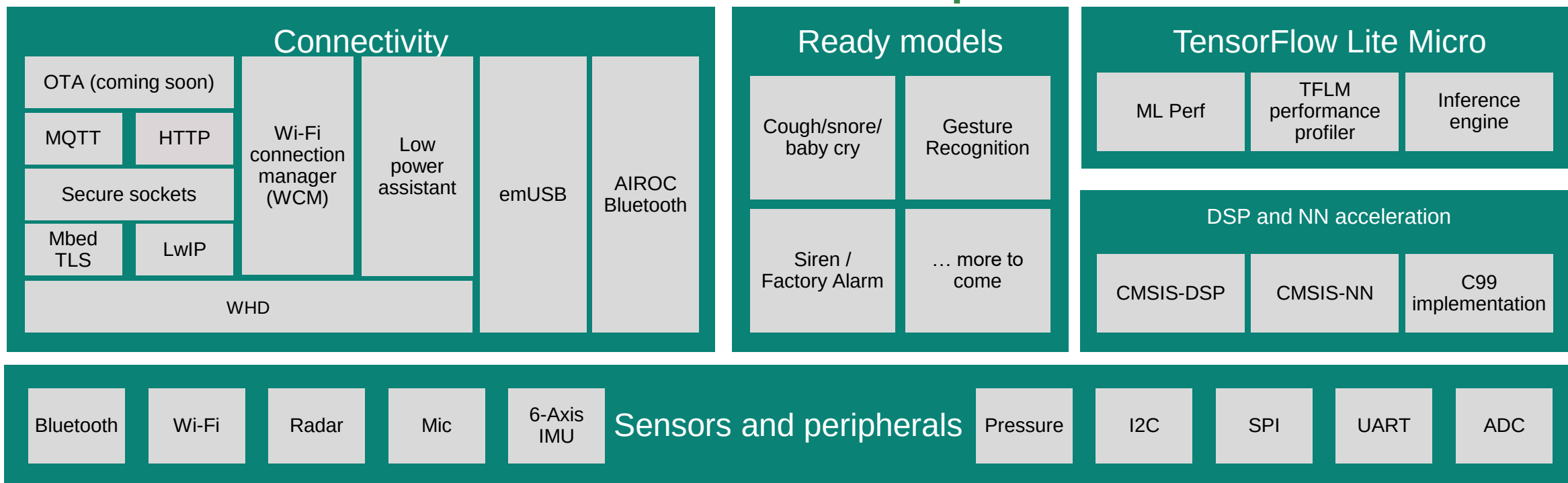
Wi-Fi, HTTP client/server

Radar presence detection

Radar gestures

Audio event detection

PSOC™ 6 software components



DEEPCRAFT™ by Imagimob Overview

/ Imagimob, An Infineon Technologies Company

Our mission is to be the best, fastest way for our customers to take their Edge AI models into production.

Imagimob in a nutshell

- **AI Expertise:** Extensive AI/ML engineering experience from 50+ customer projects.
- **An Infineon Company:** Extended support for Infineon microcontrollers and sensors. Customers can leverage Infineon's expertise for manifold applications.
- **DEEPCRAFT™ Studio:** Our state-of-the-art Edge AI/ML development platform is fully embedded into Infineon's solution ecosystem, making model development, training and optimization on target hardware easy and fast.
- **DEEPCRAFT™ Ready Models:** Production ready, off-the-shelf Edge AI models.



Lukas Wirkestrand

System Application Engineer at Imagimob
Wirkestrand.external@infineon.com

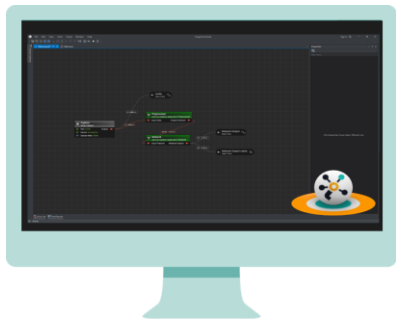
/ DEEPCRAFT™ Edge AI/ML Software enables easy integration into all designs and fast go-to-market

DEEPCRAFT™ products provide innovative, trustworthy, convenient and green AI software solutions for customer
Our portfolio offers end-to-end Edge AI and best-in-class system performance in combination with Infineon Hardware

Edge AI Development Platform

For data collection, pre-processing, model training, model conversion & deployment

DEEPCRAFT™ Studio



Edge AI Models

Ranging all the way from fully trained and ready to deploy to getting you started easily with model development. All customizable through DEEPCRAFT™ Studio

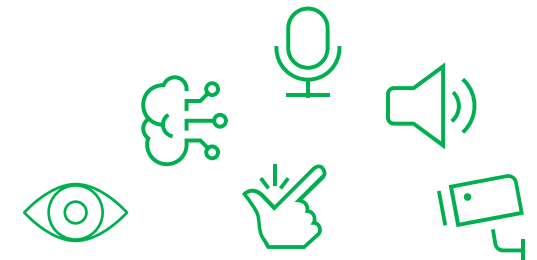
DEEPCRAFT™ Ready Models 5+
DEEPCRAFT™ Starter Models 10+



Edge AI Solutions

Collection of high value tools, models and configurators to add AI based capabilities in voice, audio, vision and sensing

DEEPCRAFT™ Voice Enhancement*
DEEPCRAFT™ Voice Assistant*
DEEPCRAFT™ Face ID*
Liquid Level Sensing



* in development

/ DEEPCRAFT™ Ready Models

AVNET

Our offering of production quality, ready to deploy Edge AI models

Available now

Baby Cry Detection 

Factory Alarm Detection 

Cough Detection 

Siren Detection 

Direction of Arrival 

Snore Detection 

Fall Detection 

Gesture Detection 

In development

Surface Detection 

Presence Detection 

Object Detection 



Audio



Radar



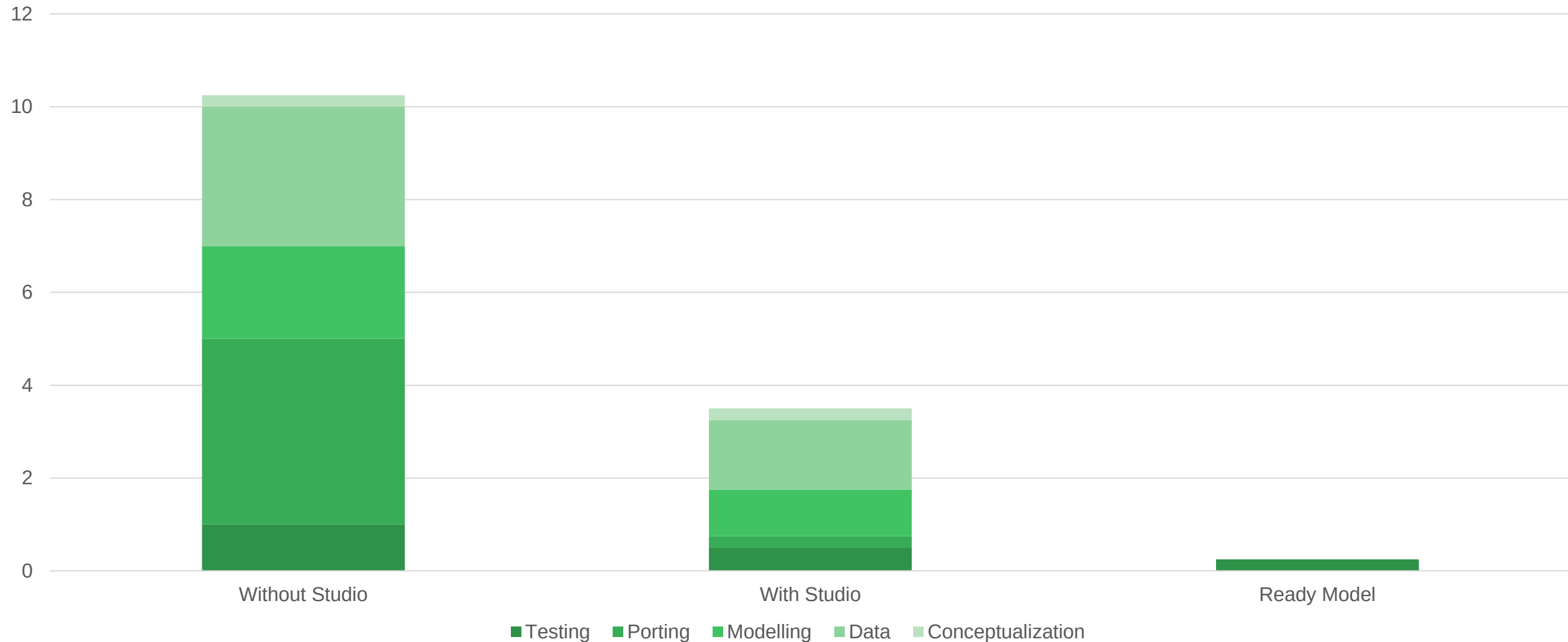
Accelerometer



Vision

/ Benefits of DEEPCRAFT™ Studio and Ready Models

Save months of developer time using optimized AI models



/ DEEPCRAFT™ Ready Model for Gesture Detection

The DEEPCRAFT™ Ready Model for Gesture Detection is a radar-based system specifically designed to detect different gestures a person performs in close proximity to a radar sensor. These include 5 different gestures; 4 swipe gestures and 1 push gesture. This model can be used in different appliances, monitor screens, laptops or other such applications where doing gestures in short range is required.



Technical specs

- Detects 5 different kinds of gestures – gestures should last 1-2 seconds
- Detects ~92% of gestures at distance of 10 cm – 70 cm from radar
- Designed to work with different heights, speeds, distances and angles
- Robust against different hand movements that are not the 5 gestures
- Can be further developed to recognize more gestures

Model specs

- Optimized to run in real-time on an Infineon PSOC™ 6 with XENSIV™ BGT60TR13C radar sensor
- Memory footprint: RAM: 3 kB, Flash: 45 kB
- Inference time: 8 ms, every 91 ms
- The model expects: range, velocity, azimuth, elevation, magnitude at 33 Hz from the radar

Possible use cases:
Contactless control of computer monitor, household appliances, etc.
Sleek, no button product design.



/ DEEPCRAFT™ Ready Model for Fall Detection

The DEEPCRAFT™ Ready Model for Fall Detection is an accelerometer-base Edge AI model designed to detecting a person falling, more specifically designed for the elderly. The model is intended for a wrist-worn device.



Technical specs

- Detects falls, with sensitivity tailored especially for the elderly.
- Detects ~93% of falls.
- Designed be very robust against False Positives (FP): 1.8 - 2.25 FP/week for more active users
- Ready to be tested on Infineon PSOC™ 6 AI Evaluation Kit

Model specs

- Optimized for running in real-time on Infineon PSOC™ 6 MCUs
- Memory footprint: RAM: 6 kB, Flash: 15 kB.
- Inference time: 1 ms, every 20 ms.
- Able to detect a fall using an accelerometer with the following sensor settings:
 - Scale = +/- 8g
 - Resolution = 12 or 16 bit
 - Sampling Frequency = 50 Hz

Possible use case:

The model could be added to a watch or smart bracelet to detect when the wearer falls and alert care takers or emergency services.



/ DEEPCRAFT™ Ready Model for Factory Alarm Detection

The DEEPCRAFT™ Ready Model for Factory Alarm Detection is an audio event detection system specifically designed for detecting alarms and sirens in a factory setting. The purpose of the model is to increase safety in factories.



Technical specs

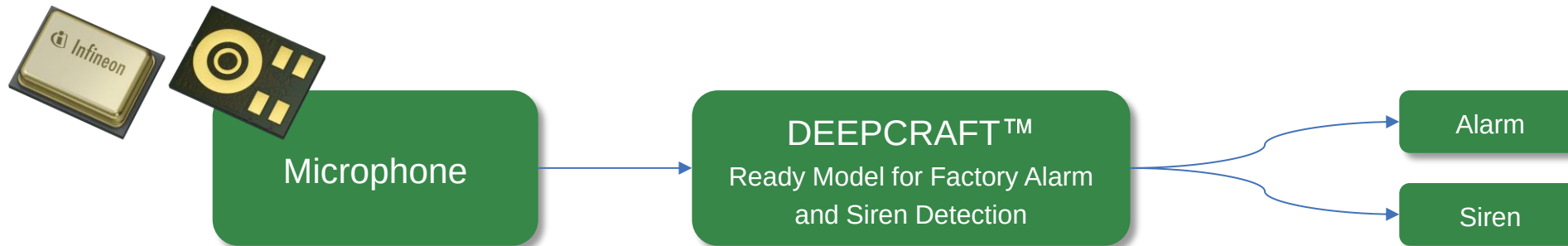
- Measures alarm and siren sounds longer than 1.2 seconds
- Detects ~90% of alarm and siren sounds in a factory setting.
- Signal To Noise Ratio must be at least 10 dB to be detected.
- Designed to be robust against different distances and angles from the sound source.
- Designed to be robust against diverse factory background sounds.

Model specs

- Optimized to run in real-time on an Infineon PSOC™ 6, and runs well within the hardware limitations.
- Memory footprint: RAM: 27 kB, Flash: 32 kB.
- Inference time: 144 ms, every 800 ms.
- Able to detect a cough from sound data with: 16000 Hz Sample Rate, 1 (Mono) Channel, 16 bit Bit Depth.

Possible use case:

This model can be used to alert workers wearing headphones or protective gear over their ears when an alarm goes off.



/ Which MCU you need for each DEEPCRAFT™ Ready Model

Ready Model	Recommended MCU*	RAM (kB)	Flash (kB)	Inference (ms) on PSOC™ 6	Sensitivity (%)
Gesture detection	PSOC 6 (CM4)	3	45	8	92
Fall detection	PSOC 4 (CM0)	6	15	1	93
Snore detection	PSOC 6 (CM4)	24	53	23	92
Baby Cry detection	PSOC 6 (CM4)	27	27	127	95
Cough detection	PSOC 6 (CM4)	27	82	150	95
Factory Alarm detection	PSOC 6 (CM4)	27	32	144	90
Siren detection	PSOC 6 (CM4)	60	51	20	92
Direction of Arrival (Audio)	PSOC Edge (U55)	826	186	82**	97

* Minimum requirement for running the model, higher processors offer better performance and scalability

** This model cannot run on PSOC™ 6, inference is instead calculated on an Arm Cortex® -A72 processor

CM4 => Arm® Cortex® -M4

CM0 => Arm® Cortex® -M0

U55 => Arm® Ethos™ -U55

/ DEEPCRAFT™ Ready Models

- Production-ready Edge AI models
- Created by experts
- Trained, tested and optimized for Infineon hardware
- No additional ML development required

Test the Ready Model for Free

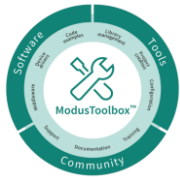


Add the Ready model to your product



Distribute your product

/IOTCONNECT™



Free /IOTCONNECT Quick Start projects available on GitHub & ModusToolbox™



[CY8CKIT-062S2-AI](#)



[CY8CKIT-062S2-43012](#)

Test and build a PoC on an Infineon evaluation kit



Sign the license agreement



Get the full library

[Contact Imagimob to Request Ready Models](#)



Pay a royalty per device

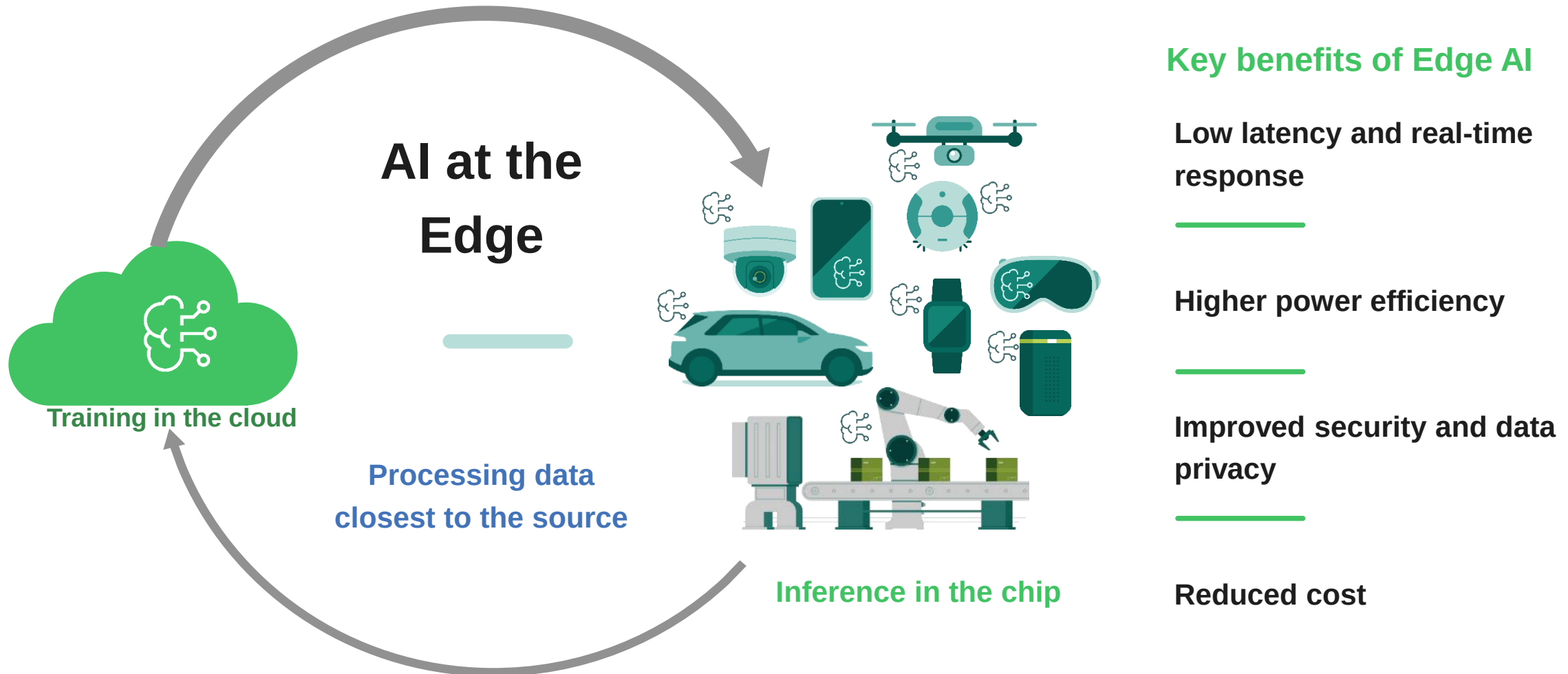


Deploy the model in your product

/IOTCONNECT Overview

Purpose for Driving Intelligence to the Edge

Real-time requirements and the need for power-efficiency, security and privacy drives AI-processing at the edge



/ Purpose for Driving Edge Intelligence to the Cloud

Data Gathering at Scale



IoT connects devices to aggregate real-world data from all nodes, creating a comprehensive dataset for training and optimization.

Continuous Model Improvement



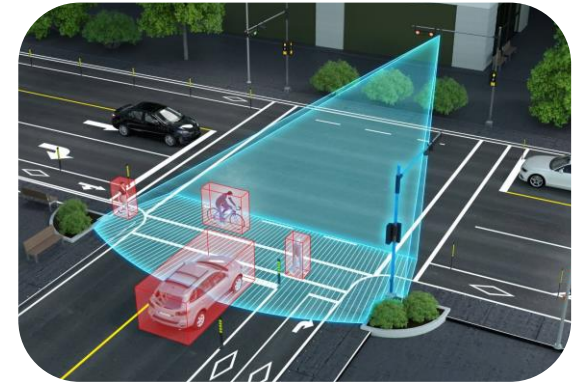
IoT enables agile model retraining and deployment, ensuring devices adapt to new conditions

Lifecycle & Security: Updates



IoT allows OTA updates for AI models and ensures compliance with evolving security standards

Unified System Intelligence

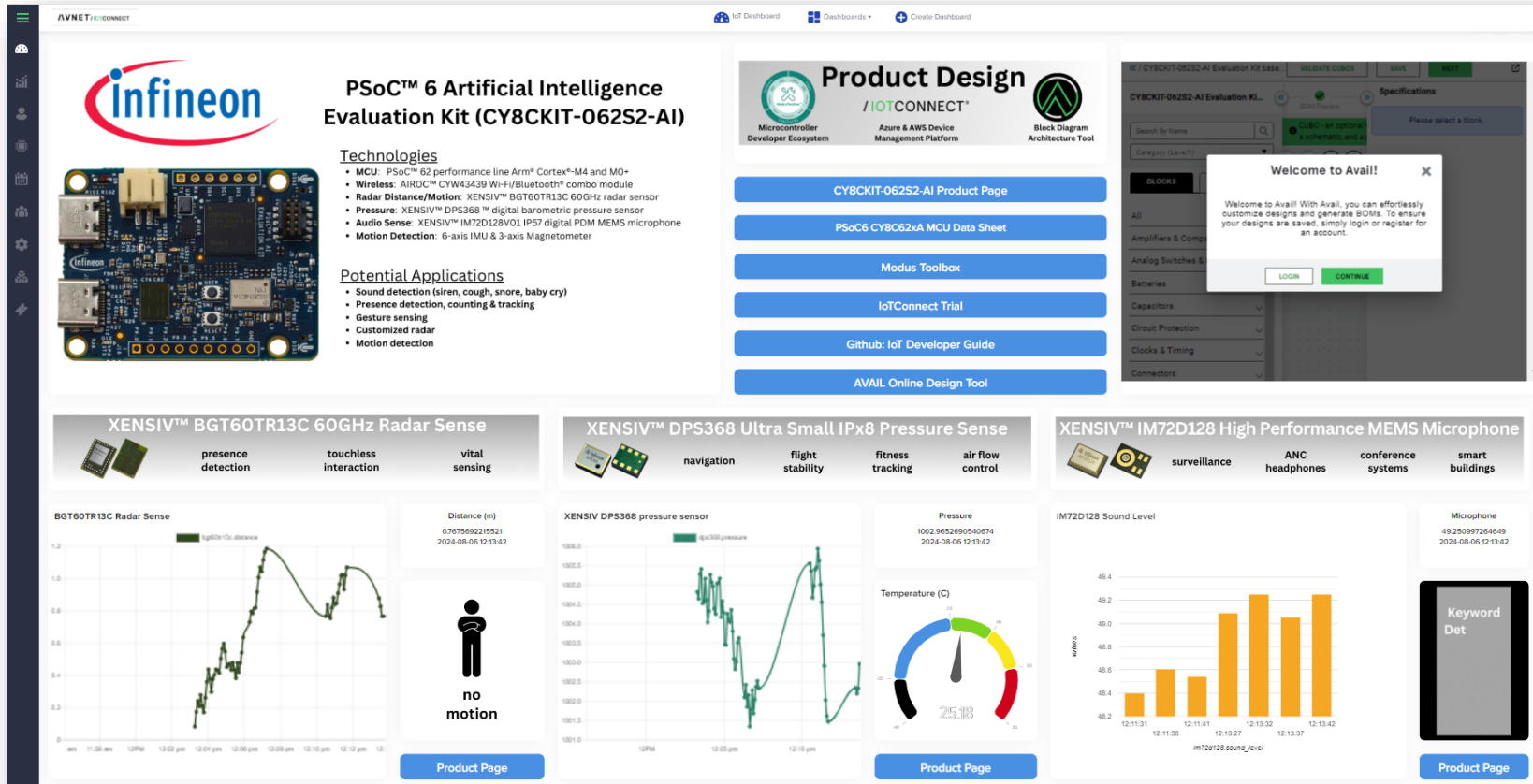


IoT connects devices into collaborative networks for smarter, global decision-making

/Inspiring AI Innovation with the PSoC™ 6 AI Kit

PSOC™ 6 AI Evaluation Kit Features:

- Compact, wireless hardware platform for machine learning innovation
- Seamless integration with Imagimob Studio and production-ready ML models
- Multi-sensor inputs (radar, acoustic, pressure, IMU) for diverse data collection
- Powerful toolchain enabling fast prototyping and real-world data capture



AI Possibilities:

- Baby Cry Detection
- Cough and Snore Detection
- Presence Detection
- Gesture-Controlled Devices
- Anomaly Detection in Machines
- Sleep Quality Monitoring
- Contactless Elevator Controls
- Security Alarm Systems
- Smart HVAC Control
- Elderly Fall Detection
- Pet Activity Monitoring
- In-Cabin Vehicle Monitoring
- Retail Foot Traffic Analytics
- Fitness and Activity Monitoring
- Environmental Hazard Detection

/ PSOC™ 6 AI Evaluation Kit

/ IOTCONNECT®

AVNET

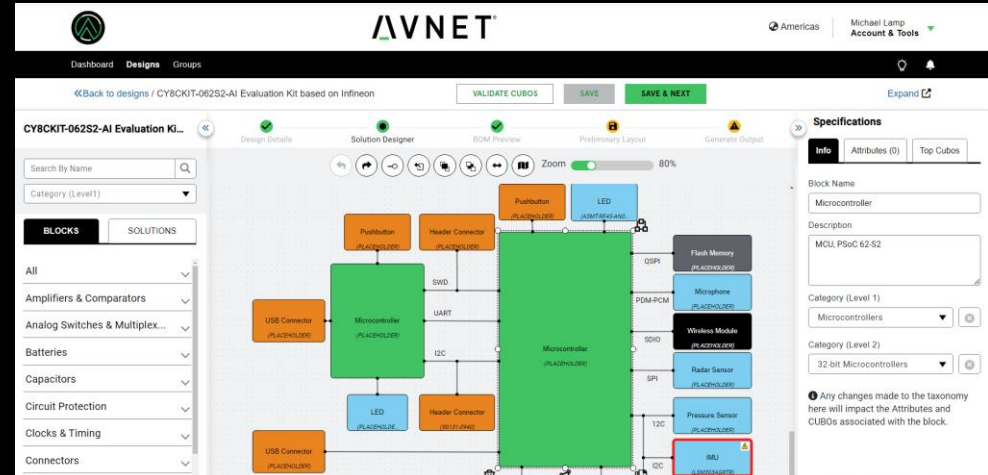
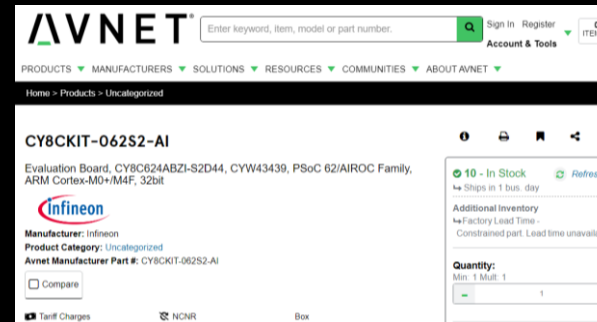
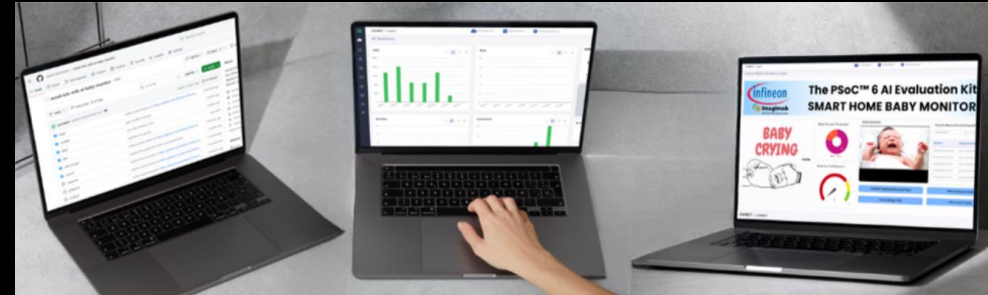
We make it easy:

- ✓ Device to Cloud onboarding in minutes
- ✓ No credit card required
- ✓ No prior Cloud experience needed
- ✓ Pre-compiled image, no additional software needed
- ✓ Demonstrate the hardware's unique functionality in the Cloud
- ✓ Enable users to build on their Quick Start journey and into development

BUY DEV KIT

IOTCONNECT FREE TRIAL

QUICK START



/ IOTCONNECT®

Quick Start

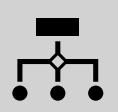
The fastest way to connect to the cloud

Watch this video to see how to connect a PSOC6 Kit to the cloud in minutes!

/IOTCONNECT: Overview

AVNET

/IOTCONNECT™
Infineon Solutions



User, Entity, and Device provisioning, management, and monitoring



Data ingestion, processing, and visualization in near real time



Supports both AWS IoT Core and Azure IoT Hub with common APIs

DESIGNED FOR . . .

Device Builders & Deployment Managers

- Device & User Mgmt
- F/W & Security Mgmt
- Dashboarding
- Internal Ops

Cloud Developers & Data Scientists

- DynamoDB / S3
- Cosmos DB / Blob
- SignalR / AWS SNS
- Event Hub / Service Bus

UX Designers & Application Developers

- RESTful API
- Predictable URLs
- Intelligent error handling
- OAuth 2.0 authentication

/IOTCONNECT: How to Get It



AWS Marketplace

SaaS Free Trial: 60-days / 500K Msgs / 10 Devices / 10 Users



Devices

IoT Core, Greengrass, LoRaWAN, Sidewalk

Connectors

Signal-R, Dynamo DB, S3, SNS, Grafana

IOTCONNECT Personal Plans

SaaS Free Trial: 30-days / 30K Msgs / 5 Devices / 5 Users

Plan	Duration	Price
Personal Plan	1 Month	\$30
Personal Plan	3 Months	\$150
Personal Plan	1 Year	\$300

Personal SaaS Plans
AWS or Azure
Messages: 30K – 5M
Devices: 5 – 100
Users: 5 - 100

Azure Marketplace

Gold Microsoft Partner

Devices

IoT Hub

Connectors

Power BI, Azure Notification Hub, Signal-R, Cosmos DB, Blob

Contact Us

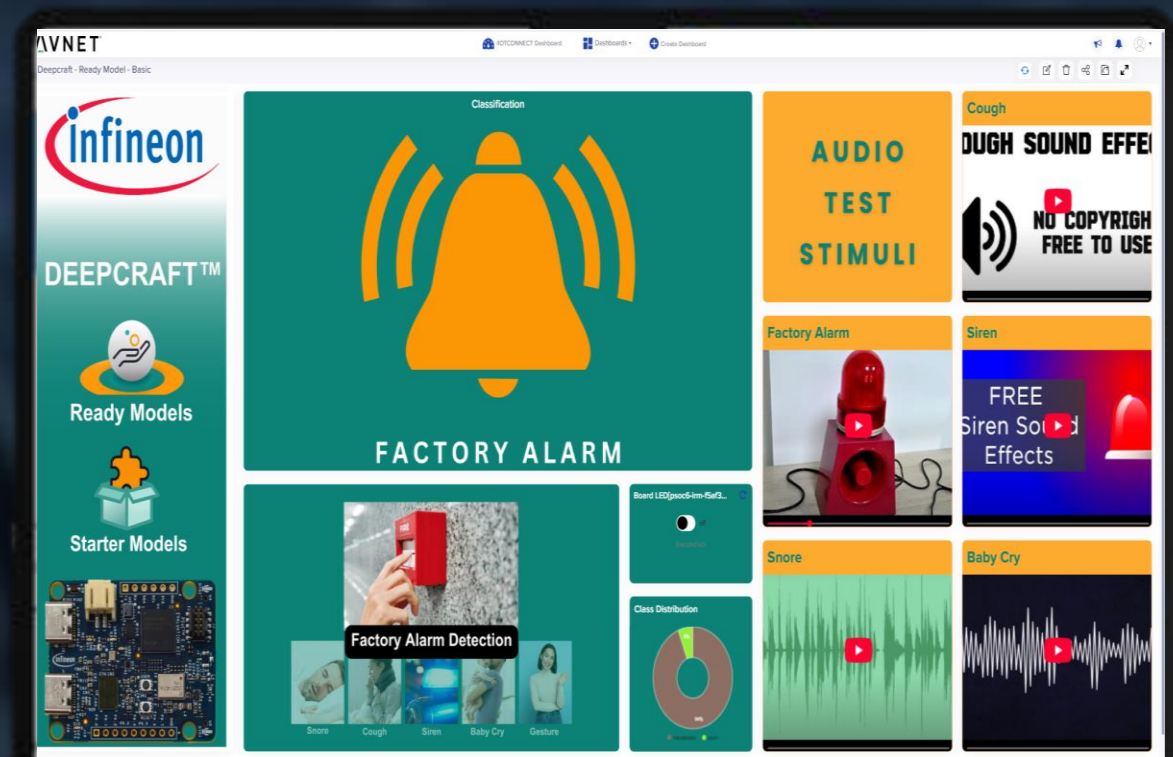
IoT@Avnet.com



- IOTCONNECT pricing and details
- AI & Data Analytics
- Enterprise Application Development
- Embedded Software & System Integration
- Global Technology Supplier Ecosystem

/ PSOC™ 6 DEEPCRAFT™ AI Demonstration

AVNET



Ready Model Example Project

[https://github.com/avnet-
iotconnect/avnet-iotc-mtb-ai-
imu-example](https://github.com/avnet-
iotconnect/avnet-iotc-mtb-ai-
imu-example)

Deploy AI devices within an enterprise AWS and Azure infrastructure

DEEPCRAFT™ AI Model Showcase

- **IMU-Based Models (Fall Detection - Elderly Care)**
 - Scenario introduction, personas involved (administrator, Emergency Response).
 - Dashboard examples with analytics (historical data, alerts).
- **Microphone-Based Models (Smart Manufacturing - Siren & Alarm Detection)**
 - Siren detection overview and technical capabilities.
 - Factory alarm detection and practical safety use cases.
- **Radar-Based Models (Gesture Detection & Retail Scenarios)**
 - Gesture detection Ready Model and practical retail use cases.
 - Radar technical specifications and transformation widgets.

Fall Detection (IMU)

[/IOTCONNECT Dashboard](#)

[Dashboards ▾](#)

[+ Create Dashboard](#)

Deepcraft - Fall_Detection

Resident Safety Control Center

Notifications

View All

Date	Resource	Event Name	Condition	Severity
Mar 28, 2025 21:21:24	fd-paul	RuleMatched	fall_detected = true	
Mar 28, 2025 21:20:26	fd-paul	RuleMatched	fall_detected = true	
Mar 28, 2025 21:18:22	fd-paul	RuleMatched	fall_detected = true	
Mar 28, 2025 20:27:21	NA	DeviceDisconnected	NA	

Resident Location

Select Device to Track ▾

Deployed Fall Detection Devices

3
Connected

0
Released

1
Disconnected

4
Total

Latest Registered Devices

Device ID	Entity	Date & Time
fd-steven	Fall Detection	March 28, 2025 00:13:58
fd-paul	Fall Detection	March 27, 2025 20:59:28
fd-mary	Fall Detection	March 27, 2025 20:38:02

Active Notifications

Rules

Last Refreshed

2025-03-28T21:25:55 [UTC]

DEEPCRAFT™

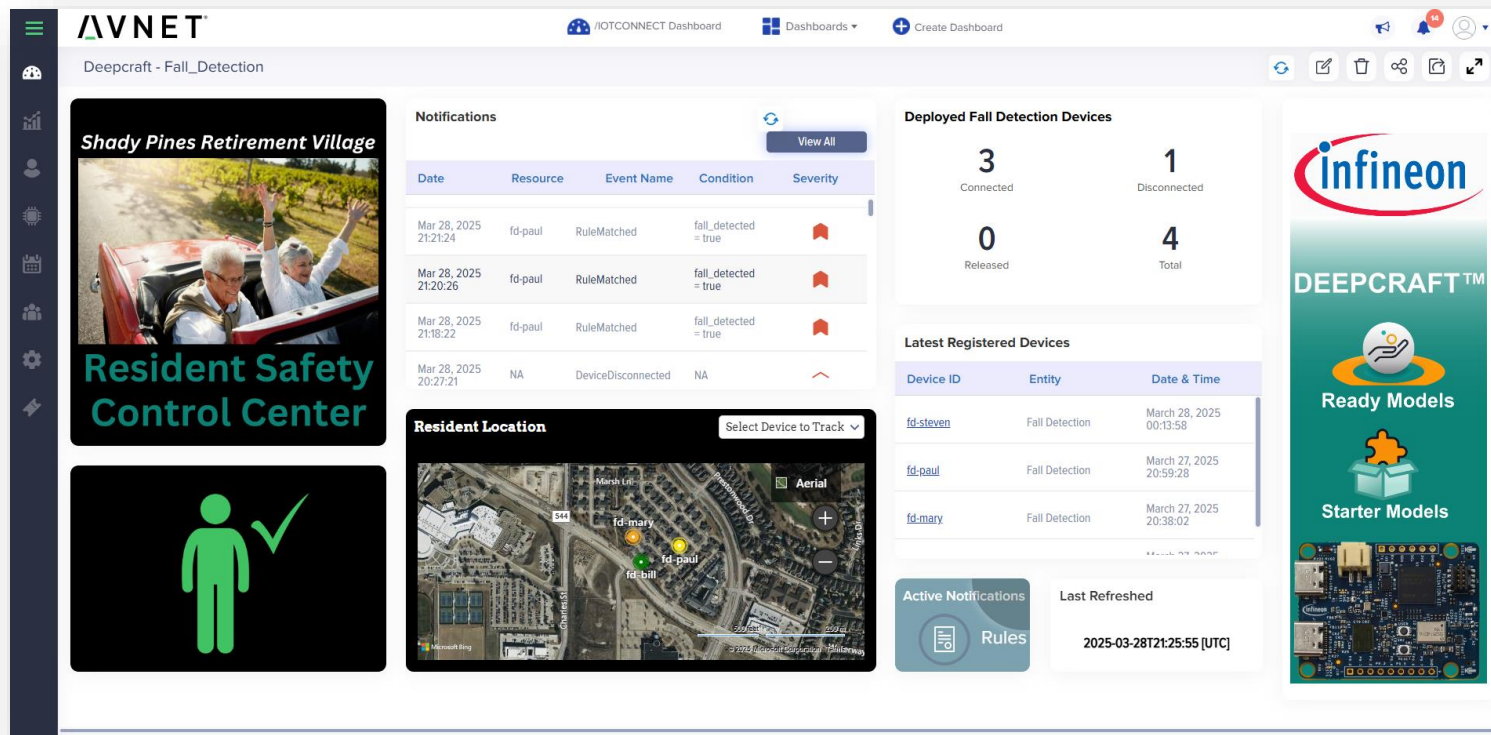
Ready Models

Starter Models

/ Fall Detection with DEEPCRAFT™ and /IOTCONNECT (IMU)

Scenario:

An assisted living facility operates a centralized control center equipped with /IOTCONNECT, monitoring multiple fall detection devices worn by residents. Each device uses Imagimob's DEEPCRAFT™ Edge AI model to accurately detect falls in real time. When a fall is detected, the device immediately transmits the incident to the /IOTCONNECT platform. The centralized dashboard then triggers alerts and notifications to caregivers instantly, allowing prompt response and assistance. The facility management team can oversee all incidents, manage devices remotely, and continuously optimize fall detection performance through easy retraining and rule adjustments, ensuring maximum safety for residents.

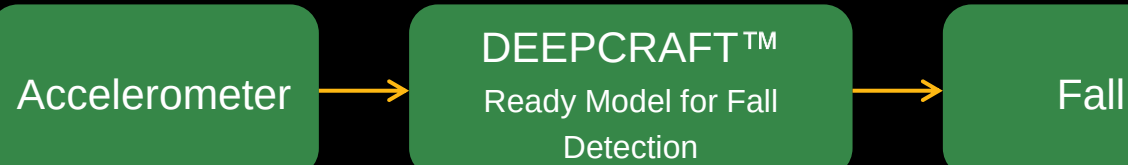


/IOTCONNECT Features:

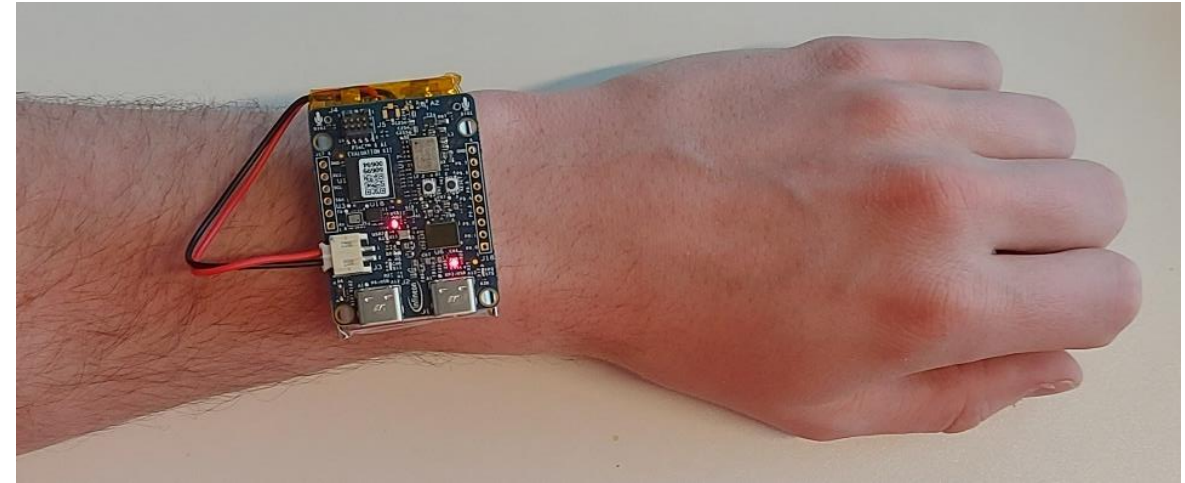
- **Real-time Alerting:** Instant notifications to caregivers when a fall is detected.
- **Centralized Device Monitoring:** Unified dashboard for tracking the status of all deployed devices.
- **Event Logging and Analysis:** Comprehensive logging of fall incidents, device activities, and triggered responses.
- **Cloud-Based Rules and Automation:** Automated triggering of alerts, responses, and escalations based on configured rules.

/ Fall Detection

AI Evaluation Kit Testing Instructions



The model is tailored to detect dangerous falls of elderly people, and it performs best on unpadded falls. However, we recommend testing with a healthy adult and placing pillows to avoid injury - the model works well in these instances too. The model is designed to have a very low False Positive (FP) per user per week; for that reason, it requires the person to lay still for 7-10 seconds after the fall.



To correctly set up the IMU orientation, make sure that the accelerometer X, Y, Z axis and values are as shown in the figures below:

Figure 1 : Bird's eye view of arm flat on table

Figure 2 : Y = 1, X, Z = 0 --- hand held up

Figure 3 : X = -1, Y, Z = 0 --- hand outstretched, palm facing front

Figure 4 : Z = 1, X, Y = 0 --- hand outstretched, palm up

Model version	Sensitivity	False positives per user per week
1.9 (current)	93.3% (28/30)	1.8-2.25
1.8	93.3% (28/30)	1.8-2.25
1.7	90% (27/30)	2.275-2.8
1.6 (baseline)	90% (54/60)	N/A

[illegible]

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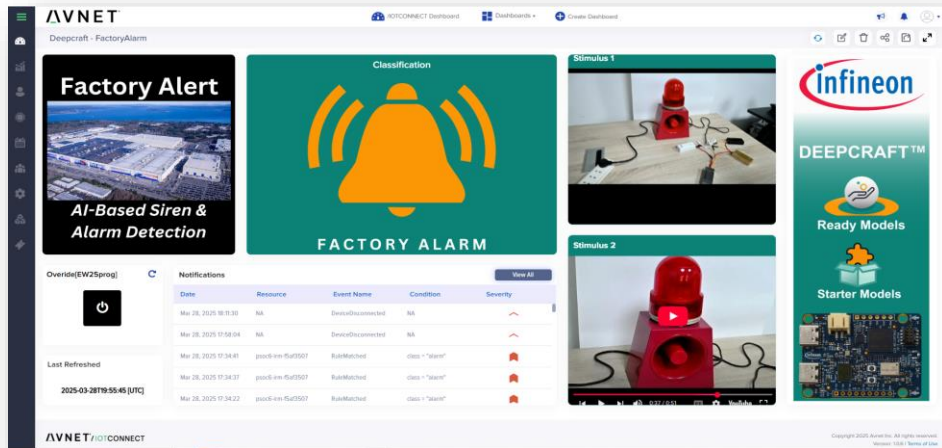
/ Smart Manufacturing: Siren Detection (Acoustic)

Real-time detection and notification of factory alarms and sirens using advanced audio-based AI Scenario:

In a large factory, various sources could trigger an alarm, such as machinery malfunctions or manually activated emergency alarms. The DEEPCRAFT™ Ready Model actively listens and accurately identifies these alarms, triggering an automated response via /IOTCONNECT cloud-based rules. When an alarm is detected, the system immediately activates a visual alert (red LED) across the facility and notifies the central control center through real-time dashboard updates.

The control center sees an immediate update via transformation widgets, indicating an active alarm. Simultaneously, /IOTCONNECT logs the event, triggers UI alerts, and can notify designated personnel through email, SMS, or other push mechanisms.

Once the situation is resolved, the control center can easily deactivate the alarm centrally using the dashboard's control interface (on/off button), disabling the alarm system and turning off the visual indicators.



/IOTCONNECT Features:

- **Real-time Notifications:** Immediate alerts when an alarm is detected
- **Centralized Device Management:** Remotely activate or deactivate alarms
- **Event Logging:** Comprehensive tracking of device statuses, events, and alarms
- **Cloud-Based Rules:** Automated response triggers upon alarm detection

AI-based sound recognition provides enhanced operational safety, response capabilities, and streamlined management in industrial environments.

/ From Sound to Action: Built on Infineon's MEMS Microphone

IM69D130 MEMS Microphone

- High-fidelity and far-field audio recording
- Matched, noise and distortion-free audio signals
- Ultra-low group delay for latency-critical applications

Features

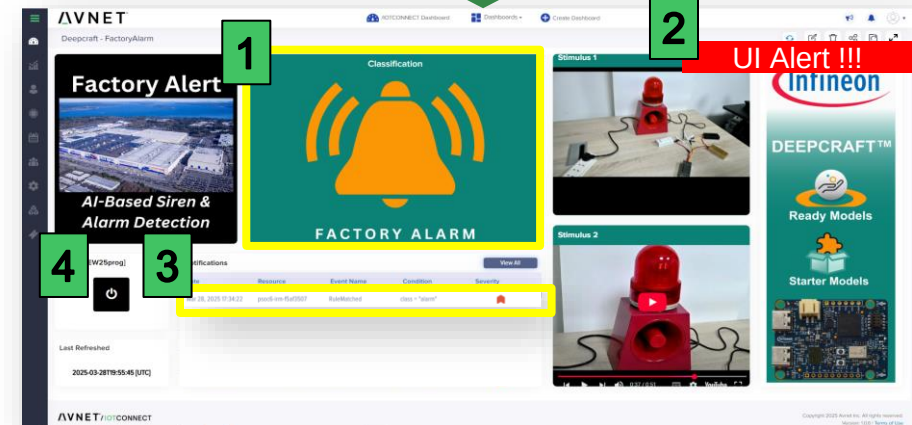
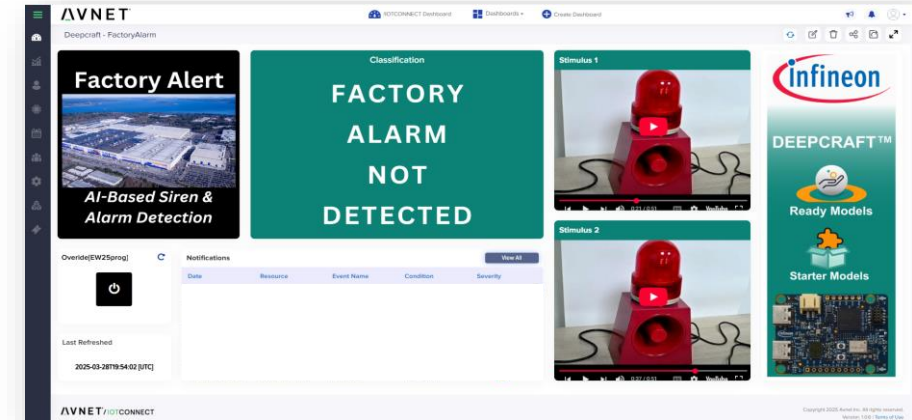
- Package: 4 mm x 3 mm x 1.2 mm
- 69 dB(A) signal-to-noise ratio
- < 1% total harmonic distortions up to 128 dB SPL
- Acoustic overload point at 130 dB SPL
- PDM interface with 6 μ s group delay at 1 kHz
- Sensitivity (-36 $\pm$ 1 dB) and phase ($\pm$ 2 deg) tolerances
- 28 Hz low-frequency roll-off

DEEPCRAFT™

Ready Model for Factory Alarm
and Siren Detection

Siren

Alarm



1. **Transformation Widget** visual alert
2. **UI Alert** across the /IOTCONNECT Portal
3. **Activated Rules** are categorized and logged in the platform
4. **Device Command** can override device alerts

/ IOTCONNECT Device Rules: C2D Commands

AVNET

AVNET /IOTCONNECT Dashboard Dashboards + Create Dashboard

Home / Overview / Device / Rules

Enter New Rule

Edit Rule

Rule name:* Factory Alarm

Template:* psoc6airm

Severity levels:* Critical

Attribute:* class

Condition:* Is equal to

Value:* alarm

Rule Applies On

☐ All Device of Selected Entity

☒ Selected Devices (Maximum 10 devices are allowed)

☐ Device Group

psoc6-irm-f...

Templates Devices Commands

AVNET /IOTCONNECT Dashboard Dashboards + Create Dashboard

Trigger C2D Command

Notification Type

DeviceCommand Email Push WebHook UI Alert UI Notification

Enable : ☒

Device Command Details

Select Command:* board-user-led

Parameter Value:* on

☐ Ignore Preference: ?

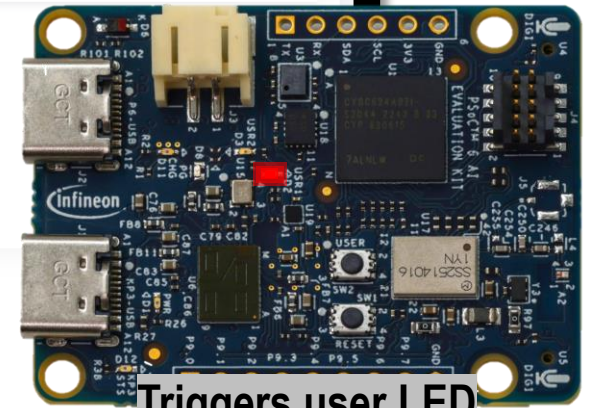
Audience

Roles: Select Roles

Users: Select Users

Michael La... x

Templates Devices Commands Rules Firmware Batch Import Data Export



Triggers user LED

IoTCONNECT Dashboard

Dashboards

Create Dashboard

Deepcraft - Gesture Recognition

Wave-to-Explore

AI-Powered Virtual Navigation

Virtual Window

Instructions

Swipe Left/Right

Rotate the view

Swipe Up/Down

Look up and down

Push (forward gesture)

Move forward

Latest Classification

not-detected

2025-03-28 15:20:54

Last Refreshed

2025-03-28T20:20:57 [UTC]

Gesture Distribution

Notifications [psoc6-irm-f5af3507]

View All

Date	Resource	Event Name	Condition	Severity
Mar 28, 2025 20:20:49	psoc6-irm-f5af...	Rule Matched	class = "SwipeDown"	☐
Mar 28, 2025 20:20:47	psoc6-irm-f5af...	Rule Matched	class = "SwipeDown"	☐
Mar 28, 2025 20:20:46	psoc6-irm-f5af...	Rule Matched	class = "SwipeDown"	☐

DEEPCRAFT™

Ready Models

Starter Models

/ Smart Retail: Gesture-Controlled Virtual Showroom (radar)

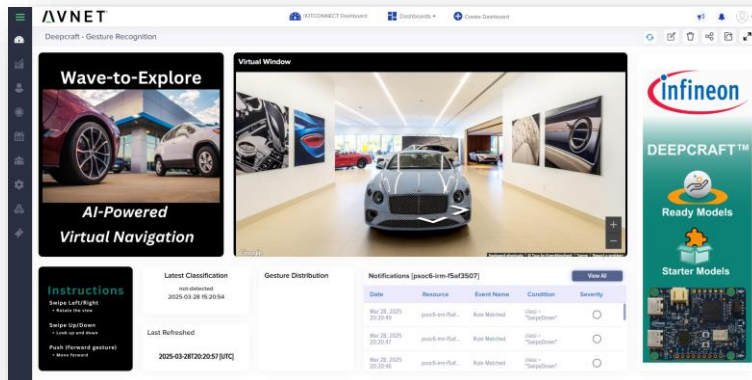
Gesture-based navigation of retail spaces using advanced AI and IoT integration

Scenario:

Today's retail customers seek engaging, interactive ways to explore products remotely or at in-store kiosks. The Imagimob DEEPCRAFT™ Gesture Recognition Model detects user gestures—swiping left/right, up/down, and pushing forward—and converts them into intuitive virtual navigation commands. Integrated with /IOTCONNECT, these gestures dynamically control a Google Street View-based virtual showroom, allowing remote exploration of products or vehicle displays.

The gesture technology functions as a standalone IoT device, placed flexibly throughout stores without requiring an attached screen. Sales personnel receive instant notifications, enabling immediate insights into customer interactions and proactive engagement with personalized recommendations.

The dashboard updates via transformation widgets, displaying recognized gestures and triggered webhooks. The platform logs interactions, remotely updates navigation start points, and delivers analytics to optimize showroom layouts and enhance customer experiences.

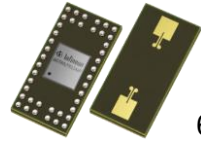


/IOTCONNECT Features:

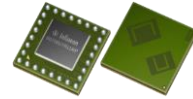
- **Remote Interaction Control:** Update views through IoT-based gestures.
- **Event Tracking and Logging:** Logging of user gestures, events, and triggered actions.
- **Sales Personnel Notifications:** Real-time alerts enabling customer engagement.
- **Cloud-Based Webhooks:** Automated cloud responses adjust showroom navigation

AI-based gesture recognition with /IOTCONNECT provides retailers with immersive customer experiences, enhanced remote selling capabilities, proactive sales interactions, and valuable insights for optimizing store environments.

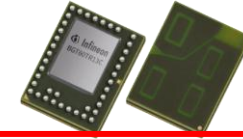
/ From Gesture to Action: Built on Infineon's mmW Radar Technology



6.7 x 3.3 mm²



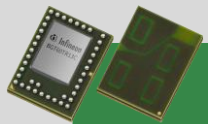
4.05 x 4.05 mm²



6.5 x 5.0 mm²

	BGT60LTR11AIP	BGT60UTR11AIP	BGT60TR13C
Positioning	Smart & cost-effective PIR replacement	Compact and cost-efficient radar chip for FMCW operations	Enable advanced radar sensing including presence, tracking and gesture control
Operating Mode	Doppler Radar with integrated detectors	FMCW (5.6 GHz BW)	FMCW (5.5 GHz BW)
Motion and Direction	Yes	Yes	Yes
Presence Detection	Only short range (~1m)	Yes	Yes
Range information	No	Yes	Yes
Angular information	No	No	Yes
Main value proposition	Completely autonomous mode	Smallest 60 GHz chip with integrated antennas	Maximum functionality and flexibility
Target use cases	Motion detection , Presence detection (short range)	Presence detection, gesture control, vital sensing, 1D ranging, material classification	Presence detection, gesture control, vital sensing, segmentation, tracking, distance

BGT60TR13C



Radar
BGT60TR13C

DEEPCRAFT™
Ready Model for Gesture
Detection

Swipe left

Swipe right

Swipe up

Swipe down

Push

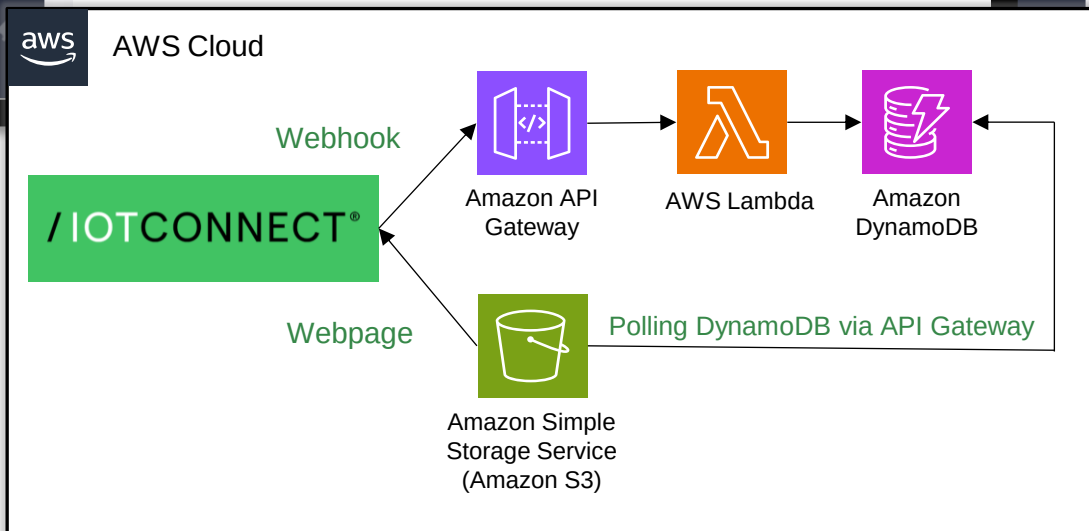
Other

/ IOTCONNECT Device Rules: Webhook Notifications

The screenshot shows the AVNET IOTCONNECT Dashboard with the 'Rules List' tab selected. The dashboard includes a sidebar with navigation icons and a top navigation bar with the AVNET logo and links to /IOTCONNECT Dashboard, Dashboards, and Create Dashboard. The main content area displays a table of rules with columns for Rule Name, Template, Entity, Rule Type, Audience Count (User and Role), Status, and Actions. A filter bar at the top right shows 'Name: swipe' with a search icon and a 'Create Rule' button.

Rule Name	Template	Entity	Rule Type	Audience Count		Status	Actions	
				User	Role			
swipe down	①	psoc6airm	Infineon	Standard	1	0	Active	
swipe left	①	psoc6airm	Infineon	Standard	1	0		
swipe push	①	psoc6airm	Infineon	Standard	1	0		
swipe right	①	psoc6airm	Infineon	Standard	1	0		
swipe up	①	psoc6airm	Infineon	Standard	1	0		

The screenshot shows the AVNET IOTCONNECT Dashboard with the 'Webhook Settings' tab selected. The settings page includes a 'Notification Type' section with tabs for DeviceCommand, Email, Push, WebHook, UI Alert, and UI Notification. The 'WebHook' tab is active, showing an 'Enable' toggle switch and 'Webhook Details' including a 'Uri' field with the URL 'https://05eimjdv0.execute-api.us-east-1.amazonaws.com/prod/setgesture', a 'Message Format' dropdown set to 'Mini', and a 'Headers' section with a key-value pair 'gesture: down'. A bottom navigation bar contains links to Templates, Devices, Commands, Rules, Firmware, Batch Import, and Data Export.



/ ModusToolbox™ Projects with IOTCONNECT

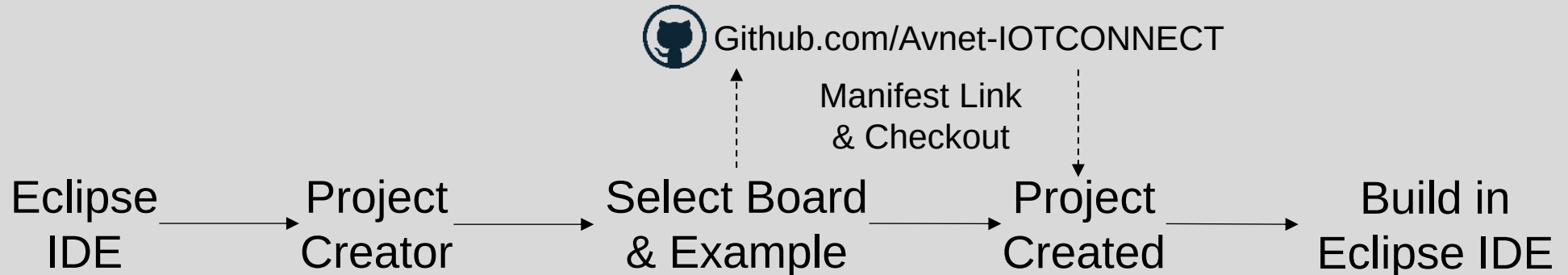
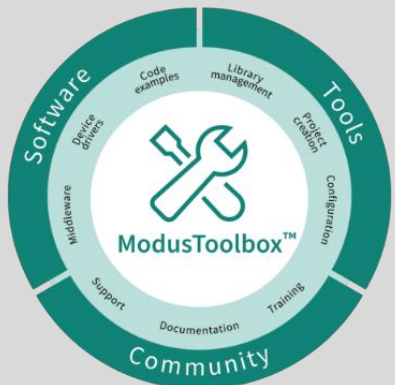
AVNET

Available in Project Creator

- [IOTCONNECT Deepcraft Ready Model](#)
- [IOTCONNECT Basic with OTA](#)
- [IOTCONNECT Optiga Trust M](#)
- [IOTCONNECT AI IMU](#)
- [IOTCONNECT AI Baby Monitor](#)

**DEEPCRAFT READY MODELS
on /IOTCONNECT.**

Available in ModusToolbox!



/IOTCONNECT Enabled Device Kits & AWS Qualification

Why Choose AWS-Qualified Devices for IoT Core?

- Qualified hardware is robust, secure, and fully compatible with AWS IoT Core.
- Devices will perform consistently in production, minimizing operational risks.
- Development and deployment timelines can be accelerated due to reduced integration complexities.
- IoT solutions will meet high reliability, security, and scalability standards required by enterprise-grade applications.



Technical Validation with AWS IoT Core Device Advisor

- MQTT protocol compliance
- Connection stability and reconnection
- Security validation (TLS, authentication)
- Message reliability and QoS
- Device Shadow functionality
- Retained message and Last Will and Testament (LWT) features
- Stress and scalability through reliability testing

/ AWS IoT Core Device Advisor Tests

MQTT Protocol Conformance Tests: Verify that devices strictly follow MQTT specifications (version 3.1.1 or 5.0)

- Ensures reliable and predictable communication.
- Confirms device compatibility with MQTT messaging patterns (publish-subscribe).
- Reduces troubleshooting time related to MQTT connectivity issues.

Connection Stability and Reconnection Tests: Ability to maintain stable connections and properly handle interruptions and reconnections

- Confirms resilience to network disruptions, ensuring device recovery and message integrity.
- Reduces downtime and improves device reliability in production environments.
- Ensures devices consistently reconnect following connection losses, preserving data integrity.

Security Validation Tests: TLS (Transport Layer Security) validation, authentication methods, and certificate management

- Ensures secure communication between devices and AWS IoT endpoints.
- Verifies correct implementation of X.509 certificate-based authentication, critical for secure device identity.
- Helps maintain compliance with security best practices, reducing vulnerabilities.

Message Reliability and Quality of Service (QoS) Tests: Confirm message delivery guarantees at different MQTT QoS levels (0, 1, and 2)

- Guarantees message delivery according to specified QoS levels, essential for mission-critical applications.
- Allows selection of the optimal QoS level based on reliability requirements versus resource consumption.

Device Shadow Tests: Validate proper interaction between devices and AWS IoT Device Shadow service

- Ensures consistent state synchronization between cloud applications and physical devices.
- Facilitates easier implementation of remote device configuration and control.
- Enhances reliability in applications involving remote monitoring and actuation.

Stress and Scalability Tests: Stress tests measure device reliability under high load or frequent connection/disconnection scenarios

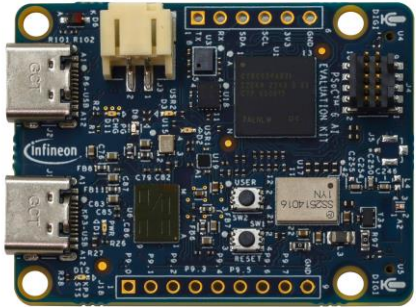
- Validates device performance under real-world, high-load conditions.
- Reduces risks associated with deployment in demanding industrial or commercial environments.
- Ensures predictable behavior at scale, critical for large-scale IoT deployments.

Retain and Last Will and Testament (LWT) Feature Tests: Confirm proper device handling of retained MQTT messages and LWT functionality

- Provides reliable device state awareness even after unexpected disconnections.
- Ensures consistent behavior in scenarios where device availability is critical, such as asset tracking or health monitoring applications.

Getting started with AI is now easier than ever – Train your first ML models with PSoC™ 6 AI Kit + DEEPCRAFT™ Starter Models!

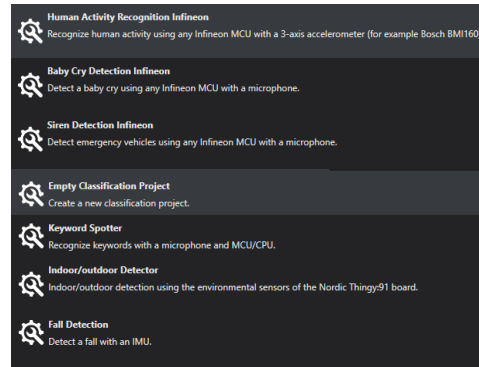
Get your
PSoC™ 6 AI Kit



CY8CKIT-062S2-AI
Scan the QR code for
more info on the kit

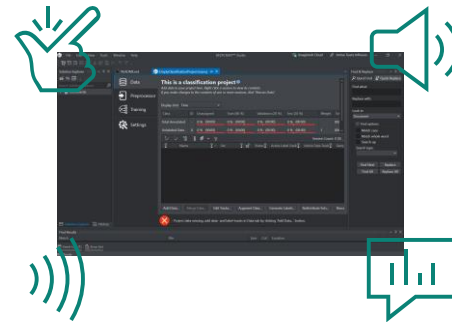


Select one of our
DEEPCRAFT™ Starter Models



... All available on GitHub and
directly accessible in
DEEPCRAFT™ Studio

Collect and add your own
data to retrain the model



... Leverage the AI Kit
and sensor extensions
as needed

Deploy & test what you
have built!



... Come back soon for
new Starter Models!

Sensors

- BGT60TR13C Radar
- IM72D128 MEMS Mic
- DPS368 Barometric Air F
- IMU from Bosch
 - BMI270
 - BMM350

PRODUCT BRIEF

CY8CKIT-062S2-AI

PSoC™ 6 Artificial Intelligence Evaluation Kit

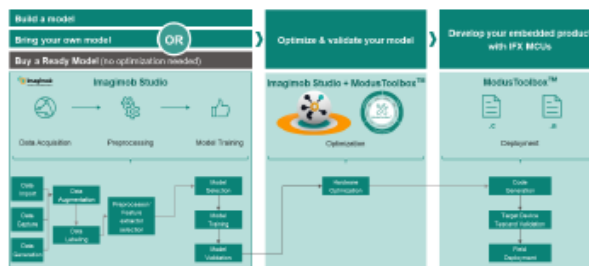
Infineon's PSoC™6 Artificial Intelligence Evaluation Kit is your perfect partner for easy prototyping and collecting real-life data to build compelling ML products fast. Being only the size of a cracker, the hardware platform focused on Edge AI enables customers to evaluate Infineon's Machine Learning platform Imagimob Studio, as well as ready to deploy ML models, and other software products.

Data collection is enabled via [PSoC™ 6 MCU](#), as well as Radar ([RG60TR13C](#)), high-performance digital MEMS microphone ([IM72D128](#)), Barometric Air Pressure Sensor ([DPS368](#)) & IMU sensors (Bosch [BME270](#), [BMM-350](#)). Connectivity is enabled with WiFi/BT BLE Combo (Murata [LBFE5K11YN](#)).



END-TO-END MACHINE LEARNING TO EMBEDDED SOFTWARE JOURNEY

No matter whether you have an ML model already and want to deploy on a target HW, or whether you want to create a new model, you can enter at any stage with Imagimob Studio.



CODE EXAMPLES (Highlights)

- **Imagimob Data Collection with IMU and Microphones:** Collect your own data and feed it into Imagimob Studio for custom model creation.
- **Imagimob Optimized Model Deployment:** Deploy your newly created model optimized and quantized for the PSoC™ 6.
- **Radar Presence Detection:** Detect whether there is somebody within range of the kit.
- **Wireless Connectivity for Cloud Connection and Reporting:** Keep the edge informed of its intelligence by synchronizing within the cloud.



Key features

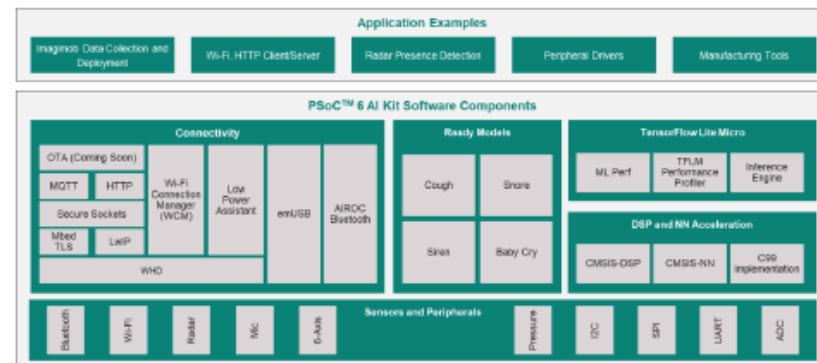
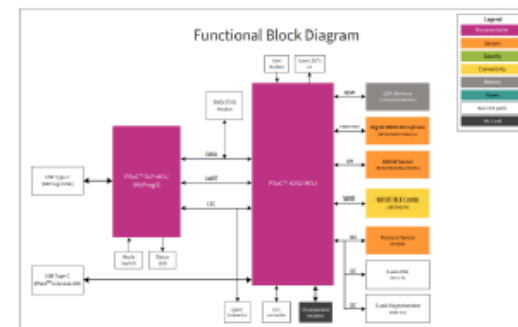
Designed for ease-of-use and end-to-end Machine Learning

- Small form-factor (35x45mm) wireless, low cost
- Solution approach with PSoC™6, QSPI Flash+ multi-sensor input (radar, microphone, pressure & 6-axis motion sensors)
- Production ready ML models: ImagimobReady Models (sitting, coughing, baby cry, snoring)
- Complete ML to embedded SW journey: ModusToolbox™ & Imagimob Studio compatible

Key benefits

- Low-cost evaluation board with efficient form factor for easy prototyping
- Rapid prototyping: Easy creation of sensor fusion-, ML-, acoustic-, time series- and radar models
- Fast time to market at minimal development cost, direct launch for PSoC™ 6 & AURIX™
- End-to-end: Collect data, create, train, evaluate & deploy ML models fast
- Seamless integration to ModusToolbox™ and Imagimob Studio with application code examples

FUNCTIONAL BLOCK DIAGRAM& SOFTWARE ARCHITECTURE



Important Links

- Find out more about the kit [here](#)
- Download the Imagimob Studio development tool [here](#)
- Evaluate one of our Ready Machine Learning Models [here](#)

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Date: 04/2024

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Scan QRcode and explore offering
www.infineon.com

Real CO₂ sensor

Photoacoustic spectroscopy (PAS) principle

Measure what matters – XENSIV™ PAS CO₂



Plug & Play for fast customer design-to-market
Infineon's quality and supply guarantee

Features

- Small form factor (14 x 13.8 x 7.5 mm³)
- 5 V power supply (IR emitter)
- SMD package delivered in tape and reel
- ±50 ppm ±5% between 400 and 3000 ppm
- Advanced compensation algorithms
- Self-calibration algorithms
- UART, I²C, PWM interfaces
- Dust-proof design (ISO 20653:2013-02)

Example: DEEPCRAFT™ Ready Model for Baby Cry Detection

The DEEPCRAFT™ Ready Model for Baby Cry Detection is an acoustic event detection system specifically designed to detect crying babies and young children up to four years old.



Technical specs

- Detects crying in babies and young children between zero to four years old.
- Designed to work at up to five meters distance.
- Detects ~95% of baby and young child cries within 10 seconds.

Model specs

- Prepared for running on an Infineon PSOC™ 6, and runs well within the hardware limitations.
- Memory footprint: RAM: 27 Kb, Flash, 26.5 kB.
- Inference time: 127 ms, every 516 ms.
- Able to detect baby or young child crying from sound data with: 16000 Hz Sample Rate, 1 (Mono) Channel, 16 Bit Depth.

Possible use case: Smart baby product or smart home device alerts adults to a baby or young child crying in another room.

Testing

- Tests showed the model works best on wailing and bawling, as compared to blubbering crying.
- Even for blubbering and other more subtle crying noises, the model accuracy is still above 90%.

Data properties

- Training was done on positive sounds, which are baby and young child cries from individuals of different genders, nationalities and ages in different indoor environments.
- Training was done on negative sounds was done to avoid triggering false positives by non-crying sounds such as pet sounds, kitchen sounds, alarms, household appliances, instruments, phones, traffic noises, adults talking and laughing, children's sounds like laughing, talking, babbling, etc., sneezing, etc.

Example: DEEPCRAFT™ Ready Model for Cough Detection

The DEEPCRAFT™ Ready Model for Cough Detection is an acoustic event detection system specifically designed for detecting a person coughing. The model is designed to detect primarily adults coughing via a wearable or other device that is nearby the user, with the purpose of identifying the user's health.



Technical specs

- Measures number of coughs per hour.
- Focuses on serious coughs, defined as those with prolonged acoustic components longer than 200 ms.
- Detects ~ 94% of coughs from between 10 cm – 2 meters distance (which means almost all coughing fits are detected when looking at long-term patterns).
- Signal To Noise Ratio must be at least 10 dB to be detected.
- Designed to misclassify no more than 6 sounds per hour.
- Designed to be robust against differences in gender, ethnicity, age, etc.
- Designed to be robust against different negative or background sounds.

Model specs

- Prepared for running on an Infineon PSOC™ 6, and runs well within the hardware.
- Memory footprint: RAM: 26.6 kB, Flash: 81.6 kB.
- Inference time: 150 ms, every 172 ms.
- Able to detect a cough from sound data with: 16000 Hz Sample Rate, 1 (Mono) Channel, 16 bit Bit Depth.

Possible use case: This model could be added to wearables in order to detect the health of the wearer.

Testing

- On-device model testing shows that the model is best used to detect multiple coughs, rather than short, single coughs. The detection increases if the cough is louder and/or closer to the device. This means that this model is most suitable for the detection of serious coughing conditions.
- The model is robust against False Positives in different real case scenarios. During testing, the model triggered only 4 in about 7 hours of non-cough sounds, namely 0.6 triggers per hour.

Data properties

- Training was done on positive sounds, which are coughs from different individuals in different environments at different distances. Wet and dry coughs were used.
- Training was done on negative sounds was done to avoid triggering false positives by non-coughing sounds such as pet sounds, kitchen sounds, alarms, household appliances, instruments, phones, vehicle noises, various weather, talking, laughing, sneezing, etc.

Example: DEEPCRAFT™ Ready Model for Siren Detection

The DEEPCRAFT™ Ready Model for Siren Detection is an acoustic event detection system specifically designed to detect emergency vehicle sirens. Rather than differentiate between various types of emergency vehicles, it accurately identifies the presence of a siren. Currently, the model is designed for use by pedestrians.



Technical specs

- Captures ~ 95% of standard US siren sounds within a proximity of 20 meters, a time frame of 3 seconds, and a 10 dB SNR.
- Designed to maintain an acceptable rate of false positives, with a max of 1 per user per day.
- Engineered to withstand variations, including the movement of emergency vehicles and the associated Doppler effect, variations in microphone angle, and changes in background noise levels due to weather conditions.

Model specs

- Prepared for running on an Infineon PSOC™ 6, and runs well within the hardware limitations.
- Memory footprint: RAM: 60 kB, Flash: 51 kB.
- Inference time: 10 - 20 ms, every 500 ms.

Testing

- As is customary in ML, we use a validation set that the model has not been trained on to validate how well the model generalizes.
- The model achieves over 90% recall on the more prominent siren sounds.
- We significantly repress false positives in non-siren sounds.
- When testing as ambulance drove by, the model did indeed trigger every time, and it did so at a distance notably longer than 50 meters (target distance).

Data properties

- Siren sounds selected to cover a range of common emergency vehicles (fire trucks, ambulance, police vehicles) with focus on, but not limited to, the US.
- Data includes sounds from different sirens, with different background noise and at different distances.
- Training on non-siren data is also done to avoid triggering false positives on sounds that aren't relevant, especially sounds that, like sirens, alternate or move across pitch, such as string trimmer, circle saw, electric shaver, cat meowing, etc.

Possible use case:
Siren noise turns off noise cancellation in headphones, making you aware of the approaching vehicle.

Example: DEEPCRAFT™ Ready Model for Snoring Detection

The DEEPCRAFT™ Ready Model for Snoring Detection is an acoustic event detection system specifically designed to detect a person snoring. The model is designed to detect primarily adults snoring via a wearable or other device that is nearby the user, with the purpose of identifying the user's health.



Technical specs

- Detects snore within 2 mins of event happening.
- Detects ~96% of coughs from between 10 cm – 2 meters distance.
- Designed to misclassify maximum 6 sounds per hour.
- Designed to be robust against differences in gender, ethnicity, age, etc.
- Designed to be robust against different negative or background sounds.
- Intended to be engaged only when the user is asleep.

Model specs

- Prepared for running on an Infineon PSOC™ 6, and runs well within the hardware limitations.
- Memory footprint: RAM: 24.3 kB, Flash: 53 kB.
- Inference time: 23 ms, every 62 ms.
- Able to detect a snore from sound data with: 16000 Hz Sample Rate, 1 (Mono) Channel, 16 bit Bit Depth.

Possible use case: This model could be added to wearables in order to detect the health of the wearer or to trigger a smart alarm clock.

Testing

- Model testing indicated that it's very robust against different kinds of background sounds such as fans or air-conditioning or people talking
- It also has minimal triggers on regular sleeping noises like breathing etc.

Data properties

- Training was done on positive sounds from different sources and was tested on individuals from around the world of different ages and ethnicities.
- Training was done on negative sounds was done to avoid triggering false positives by non-snoring sounds such as pet sounds, kitchen sounds, alarms, household appliances, instruments, phones, vehicle noises, various weather, talking, laughing, sneezing, etc.

Avnet Engineering Services offered by Softweb Solutions

AVNET

Edge AI Technologies We Cover

Security protocols:

Encryption, secure boot, trusted execution environments

Edge computing platforms:

Infineon, Renesas, Texas Instruments, STM32, NVIDIA, Intel, Google Coral

IoT integration:

Collect, process, and analyze data in real-time

Machine learning frameworks:

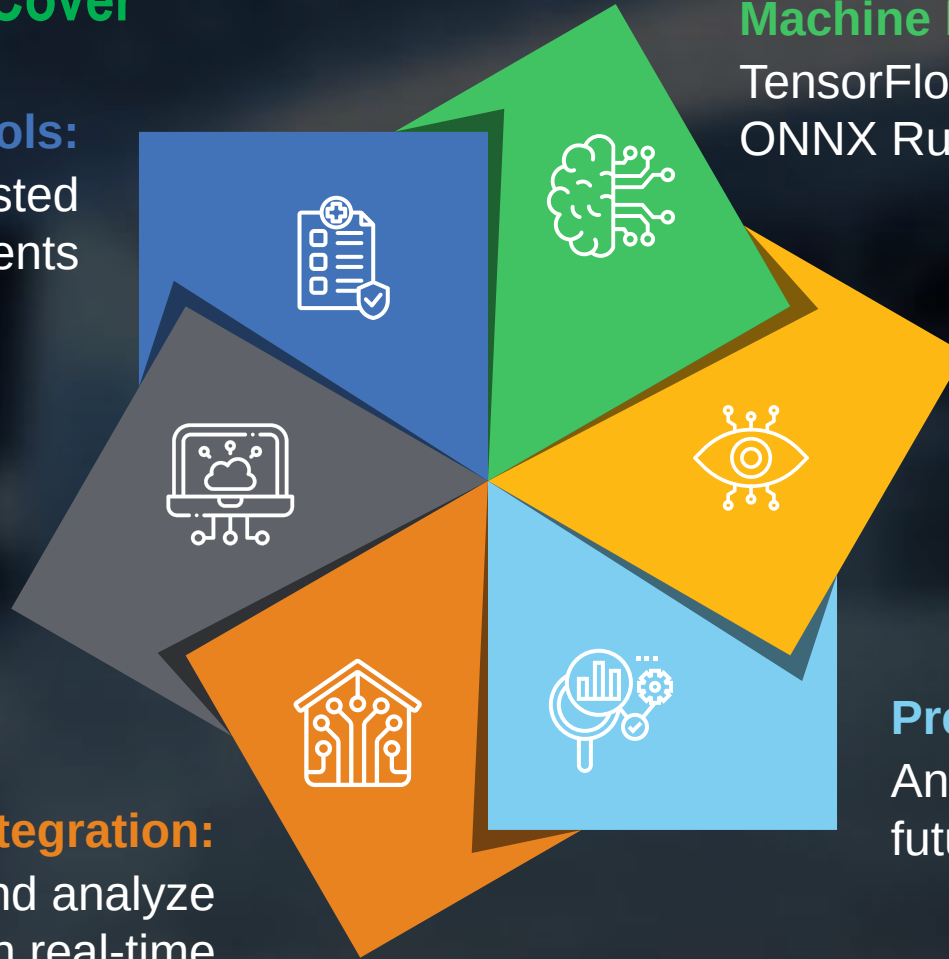
TensorFlow Lite, PyTorch Mobile, ONNX Runtime, STM32Cube.ai, SLMs

Computer vision:

Real-time visual inspection, object detection, quality control

Predictive analytics:

Anticipate and respond to future events



/ EdgeAI Services

EdgeAI Development Services

- Design, develop, and deploy AI models directly on edge devices for real-time processing.
- Utilize supplier-provided tools and resources, including SDKs and hardware-optimized libraries, to ensure seamless integration and performance.
- Support for both large, high-capacity devices (e.g., industrial servers) and embedded devices like **microcontrollers** and **microprocessors** with limited space and processing power.
- Custom model training for tasks like object detection, image recognition, and anomaly detection.

AI Model Optimization

- Optimize AI models for edge deployment by reducing size and increasing efficiency.
- Implement quantization and pruning techniques to ensure smooth operations on resource-constrained devices.

Integration with IoT and Edge Devices

- Seamless integration with IoT platforms for end-to-end edge intelligence solutions.
- Enable real-time insights by connecting AI models with sensors, cameras, and other edge hardware.

Consulting Services

- End-to-end EdgeAI consulting, from strategy development to implementation.
- Industry-specific expertise to address unique challenges in manufacturing, healthcare, retail, and more.
- Continuous post-deployment support and optimization.

PSoC™ edge portfolio

Next generation MCU-based edge device platform

Next generation

Scalable, compatible,
“future proof”*
platform for your next-
gen MCU based smart
devices

Broad MCU based SoC family



PSOC™ edge E81

- Higher performance
- More memory
- More peripheral support
- Higher integration
- Lower power



PSOC™ edge E83

Adds:

- Advanced edge AI/ML
- Audio/voice
- Vision
- Presence/gesture/motion



PSOC™ edge E84

Adds:

- Graphics
- Even more memory

– Base edge AI/ML

– Robust security

100% pin compatibility

System/software compatibility

*Ask us for industry research and OEM statements indicating rapid emergence of edge ML and robust security requirements.

/IOTCONNECT® PSOC™ 6 AI Evaluation Kit

AVNET

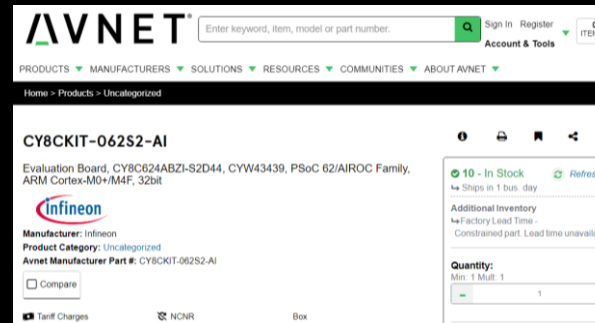
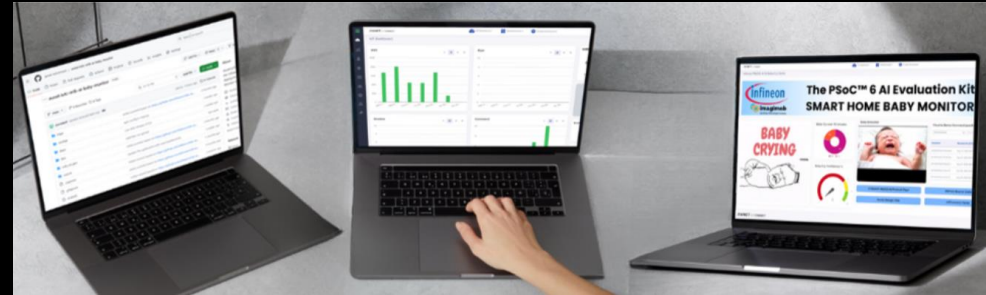
We make it easy:

- ✓ Device to Cloud onboarding in minutes
- ✓ No credit card required
- ✓ No prior Cloud experience needed
- ✓ Pre-compiled image, no additional software needed
- ✓ Demonstrate the hardware's unique functionality in the Cloud
- ✓ Enable users to build on their Quick Start journey and into development

BUY DEV KIT

IOTCONNECT FREE TRIAL

QUICK START

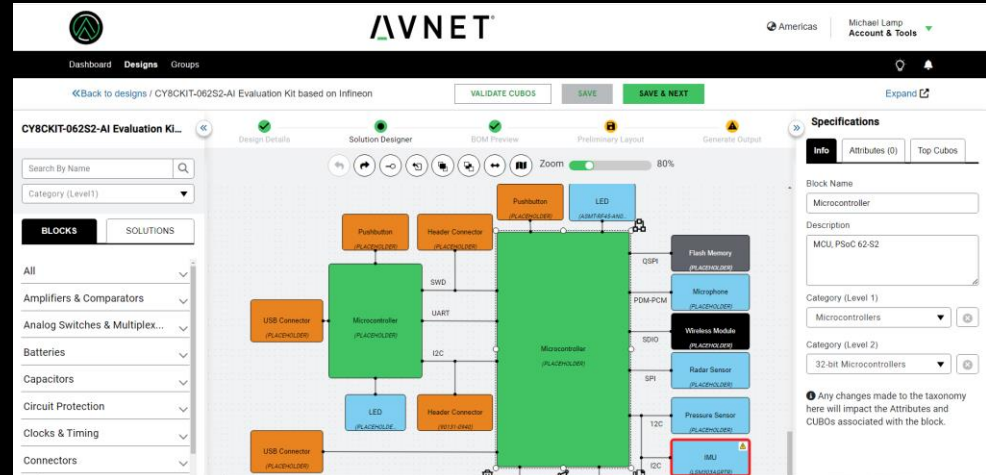


/IOTCONNECT®

Quick Start

The fastest way to connect to the cloud

Watch this video to see how to connect a PSOC6 Kit to the cloud in minutes!



/ PSoC™ 6 Imagimob AI Demonstration

AVNET



Baby Cry Detection

<https://github.com/avnet-iotconnect/avnet-iotc-mtb-ai-imu-example>

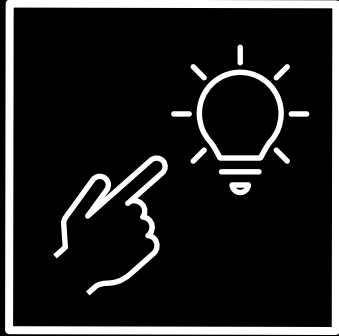


Human Activity Classification

<https://github.com/avnet-iotconnect/avnet-iotc-mtb-ai-imu-example>

Deploy AI devices within an enterprise AWS and Azure infrastructure

/IOTCONNECT®



Simple

/IOTCONNECT®
Manage

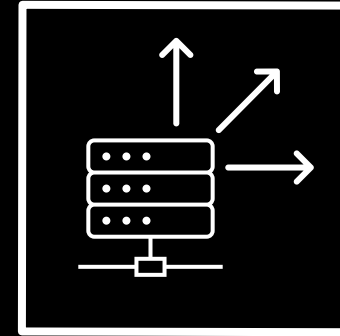
Device Management and
dashboarding made easy



Secure

/IOTCONNECT®
Secure

Robust security platform, ensuring
encrypted data



Scalable

/IOTCONNECT®
Deploy

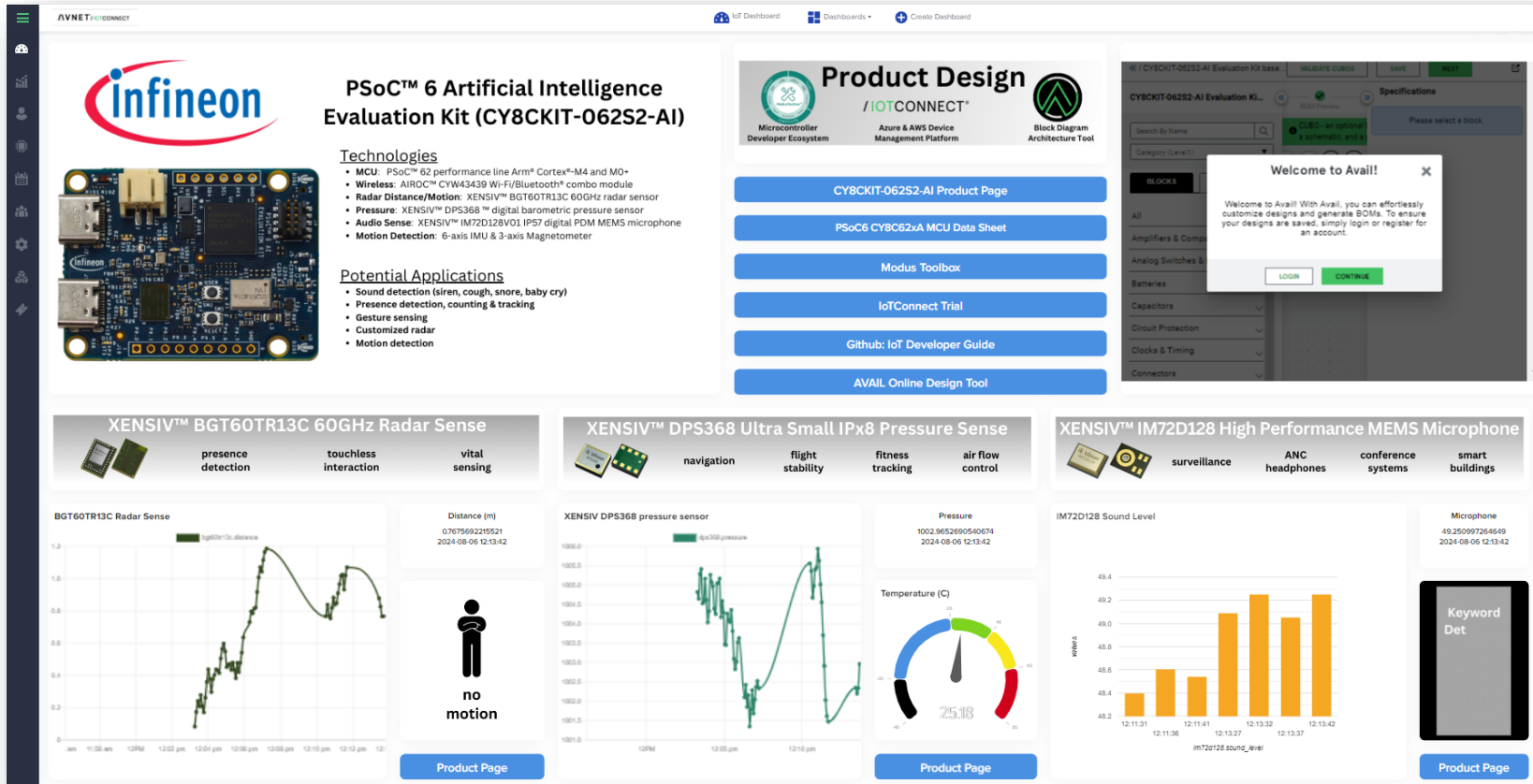
Rapidly deploy solutions to
any number of devices

IOTCONNECT makes IoT solution development simple, secure, and scalable.

Inspiring AI Innovation with the PSOC™ 6 AI Kit

PSOC™ 6 AI Evaluation Kit Features:

- Compact, wireless hardware platform for machine learning innovation
- Seamless integration with Imagimob Studio and production-ready ML models
- Multi-sensor inputs (radar, acoustic, pressure, IMU) for diverse data collection
- Powerful toolchain enabling fast prototyping and real-world data capture



AI Possibilities:

- Baby Cry Detection
- Cough and Snore Detection
- Presence Detection
- Gesture-Controlled Devices
- Anomaly Detection in Machines
- Sleep Quality Monitoring
- Contactless Elevator Controls
- Security Alarm Systems
- Smart HVAC Control
- Elderly Fall Detection
- Pet Activity Monitoring
- In-Cabin Vehicle Monitoring
- Retail Foot Traffic Analytics
- Fitness and Activity Monitoring
- Environmental Hazard Detection